

TIFFANY LIOU

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Updated 3/3/2026

EDUCATION

M.S. Physics, Northwestern University, Evanston, IL Dec 2025

B.S. Physics, University of California – San Diego, La Jolla, CA Jun 2024

TECHNICAL SKILLS

- *Computer Languages*: Python, MATLAB, LaTeX, Mathematica, HTML
- *Skills*: Code development, Bayesian statistics, machine learning, data cleaning, data analysis, academic writing

RESEARCH EXPERIENCE

Galactic Outflow Properties at Cosmic Noon, *Master's Thesis* Sept 2024 – Dec 2025

- *Advisors*: Dr. Xinfeng Xu, Dr. Allison Strom (Northwestern CIERA)
- Constrained galactic outflow scaling relations using 98 Cosmic Noon galaxies from the Keck Baryonic Structure Survey and the Keck Lyman Continuum Survey, creating the largest outflow sample at $z \sim 2-3$ to date
- Built a Python pipeline for outflow detection by fitting double-Gaussians to strong nebular emission lines

Rest-UV Properties of MUSE HUDF DR2 Galaxies, *Society of Physics Students Intern* Jun 2023 – Aug 2023

- *Advisor*: Dr. Nimisha Kumari (Space Telescope Science Institute)
- Analysed scaling relations of 98 Lyman-alpha emitter galaxies at $z \sim 2-4$ from the MUSE Hubble Ultra Deep Field
- Constructed Python code that utilizes the MUSE Python Data Analysis Framework package to fit spectra and select galaxy candidates

MCMC Inference of Dark Matter Subhalo Impacts, *Undergraduate Honors Thesis* Nov 2022 – Jun 2024

- *Advisor*: Dr. Tongyan Lin (UC San Diego)
- Simulated stellar streams and perturber dark matter subhalos using the Gala Python package
- Developed Python scripts to infer the properties of dark matter subhalos using Markov Chain Monte Carlo (MCMC) methods and successfully inferred the mass, size, and 6-D kinematic information of the perturber.

Measuring Dark Matter with Stellar Streams, *CASSI Intern* Jun 2022 – Aug 2022

- *Advisor*: Dr. Josh Simon (Carnegie Observatories)
- Implemented Python statistical analysis tools to analyze the kinematic variations of simulated perturbed stellar streams and measure the impact of dark matter subhalos

PUBLICATIONS

- **Tiffany Liou**, Xinfeng Xu, Allison L. Strom, Nathalie A. Korhonen Cuestas, Claude-André Faucher-Giguère, Tim B. Miller, Gwen Rudie, Ryan F. Trainor, Naveen A. Reddy, Charles C. Steidel, Yuguang Chen, “Keck/MOSFIRE Constraints of Cosmic Noon Outflows: Velocity, Mass Loss, and Redshift Evolution” (*in prep*)
- Sara Morrissey, **Tiffany Liou**, Adam Burgasser, “Spectral Model Fitting of Cold and Distant Brown Dwarfs Detected in a Deep Survey with the James Webb Space Telescope,” 2025, BAAS, 57, 2
- **Tiffany Liou** & Nimisha Kumari, “Rest-UV Properties of MUSE DR2 Galaxies,” 2024, BAAS, 56, 2
- The SPS Observer: Lyman-Alpha Emitters Offer a Peek into Galactic Evolution, 2024
- **Tiffany Liou** & Josh Simon, “Measuring Dark Matter with Stellar Streams,” 2023, BAAS, 55, 2

AWARDS

Chambliss Award, 243rd American Astronomical Society Meeting Jan 2024

Physical Sciences Dean's Undergraduate Awards for Excellence, UC San Diego, Jan 2024

TEACHING EXPERIENCE

Teaching Assistant, Northwestern University, Evanston, IL Sept 2024 – Dec 2025

- Facilitated undergraduate physics laboratory sessions and enhanced student learning through group discussions and hands-on experiments
- Courses supported: Mechanics (Phys 131–A), Electricity and Magnetism (Phys 131–B), Modern Physics (Phys 131–C)

Supplemental Instruction Leader, UC San Diego, La Jolla, CA Jan 2022 – Jun 2024

- Designed and facilitated weekly 80-minute in-person and virtual topic sessions for undergraduate physics students, supporting over 200 unique individuals with a session return rate of 83%
- Fostered active learning environments through implementing pedagogical principles of collaboration, scaffolding, and redirection to promote problem-solving skills
- Courses supported: Mechanics (PHYS 2A, 4A), Electricity and Magnetism (PHYS 2B, 4C), Thermodynamics & Fluids (PHYS 4B), Optics & Modern Physics (PHYS 4C)

Physics Team Course Leader (Sept 2022 – Jun 2024)

- Bridged communication between the Supplemental Instruction Program Coordinators and the Physics Team through leading weekly team meetings and trainings
- Conducted session observations and ensured the quality of academic support by effectively providing constructive feedback and resources to the session leaders

Learning Assistant, UC San Diego Department of Physics, La Jolla, CA Sept 2023 – Mar 2024

- Collaborate with graduate student teaching assistants to lead discussion sessions for PHYS 2A (Mechanics) and PHYS 2B (Electricity and Magnetism)
- Deepened student problem-solving and critical-thinking skills by breaking down complex problems and guiding students to create their own solutions

PRESENTATIONS

Galactic Outflow Properties at Cosmic Noon Dec 2025

Chen Group Meeting, University of Chicago

Stellar Halo Group Meeting, University of Michigan

Harder, Better, Faster, Stronger – Warm-ionized Outflows at Cosmic Noon Oct 2025

Northwestern Physics and Astronomy Graduate Student Council PAE CRS Seminar

Rest-UV Properties of MUSE HUDF DR2 Galaxies

iPoster: 243rd American Astronomical Society Meeting

SPS Symposium 2023

Jan 2024

Aug 2023

Measuring Dark Matter with Stellar Streams

Rubin Observatories Stellar Stream Group

iPoster: 241st American Astronomical Society Meeting

CASSI Symposium

May 2023

Jan 2023

MENTORSHIP AND LEADERSHIP EXPERIENCE

STARTastro Graduate Mentor, UC San Diego, La Jolla, CA Jul 2024 – Aug 2024

- Mentored 8 students from the San Diego area community colleges through the STARTastro summer program to kickstart their academic careers in physics and astronomy
- Designed and led mentorship sessions on imposter syndrome, sense of belonging, and resources for transfer students

CASSI Alumni Mentor, Carnegie Observatories, Pasadena, CA

Jun 2023 – Present

- Served as a peer mentor for summer interns, providing both scientific and career guidance

- President, *Society of Physics Students at UC San Diego***, La Jolla, CA Mar 2023 – Jun 2024
- Led a team of 25 students to promote student interest in physics through event planning, including: scientific outreach, student talks, community town halls, and chats with speakers within physics and astronomy.

OUTREACH

Letters to a Pre-Scientist	Aug 2025 - Present
CIERA Astronomer Evenings, <i>Northwestern University</i> , Evanston, IL	May 2025
Barrio Logan Science Expo, <i>UCSD Department of Physics</i> , San Diego, CA	Mar 2024
Preuss School UCSD Science Expo, <i>Society of Physics Students at UC San Diego</i> , La Jolla, CA	Mar 2024
Astronomy on the Mall, <i>Society of Physics Students</i> , Washington D.C,	Jun 2023
Petco Park Science Expo, <i>UCSD Department of Physics</i> , San Diego, CA	Mar 2023
Outreach Team, <i>Society of Physics Students at UC San Diego</i> , La Jolla, CA	2022-2023